

Unit Hackathon

ESA BIC Czech Republic Challenge

Urban Heat & Green Cooling Using Copernicus CDSE Data

Target audience: FIT & FEL students and faculty (ČVUT).

Duration: 8 hours.

Goal: working prototype + short demo.

1. Challenge overview

Urban areas often run hotter than surrounding regions (urban heat island effect). Teams will use open satellite data accessed via the **Copernicus Data Space Ecosystem (CDSE)** to identify heat hotspots and explore how vegetation helps cool cities.

2. Task definition

- Use Copernicus datasets accessed through CDSE.
- Produce at least one **heat-related** output (map, chart, or statistic).
- Produce at least one **vegetation-related** output (NDVI map, vegetation mask, or statistic).
- Combine them into a clear insight (e.g., “areas with low NDVI tend to be hotter”).

Suggested study area: Prague (preferred) or any Central European city/region.

3. Datasets (CDSE)

Required: Use at least one dataset below (you may use both).

- **Sentinel-3 SLSTR** (Land Surface Temperature context; good for heat island signals).
- **Sentinel-2 L2A** (Multispectral imagery; compute NDVI or use pre-made NDVI visualisation layers in Copernicus Browser).

Where to find and explore the data

- [CDSE portal](#)
- [Copernicus Browser \(quick exploration & NDVI visualisation\)](#)
- [CDSE documentation](#)
- [Copernicus Land Monitoring Service \(optional layers\)](#)

4. Recommended programming language

Use Python (recommended for fastest progress and best geospatial tooling).

- **Core:** numpy, pandas, matplotlib
- **Raster/EO:** rasterio, xarray
- **Maps:** folium or plotly (optional)
- **CDSE-friendly tooling:** sentinelhub-py (if using Sentinel Hub APIs through CDSE)

[Sentinel Hub Python package docs \(examples included\)](#)

5. Getting started fast (accounts + access)

Create a free CDSE account, then explore the area and time period in Copernicus Browser. For coding,

you can start either by (A) downloading data products or (B) using APIs / notebooks.

- [Create CDSE account](#)
- [CDSE videos hub \(official playlists\)](#)

6. Working YouTube starters (verified links)

Pick **one** video below and start building. These are stable, official or widely used references.

- [A hands-on walkthrough of CDSE \(webinar\)](#)
- [Short guide to Copernicus Browser \(official\)](#)
- [CDSE website + Copernicus Browser walkthrough](#)
- [Intro to CDSE JupyterLab access \(official\)](#)
- [CDSE openEO API webinar \(optional path\)](#)

7. Deliverables

- A working prototype (map/dashboard/notebook) showing heat + vegetation outputs.
- A 2-3 minute demo explaining: data used, processing steps, and the main insight.
- A short README (how to run).

8. Evaluation (simple)

- Correct and meaningful use of CDSE Copernicus data
- Technical implementation (data pipeline + code quality)
- Clarity of insight and visualization
- Working demo

One-sentence prompt: Use Copernicus CDSE satellite data to identify urban heat islands and quantify

how vegetation cools the city.